Appendix 4. Overview of characteristics and results of studies reporting on amount and type of physical activity.

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Abe et al. 2014	Japan	THA HRA	Case-control study	Patients: 608 Age: 62 (36-98) 14% male BMI: 23 (14-34) 32% bilateral Surgery: 90% THA, 10% HRA	4.8 (2.3-7.8) years	Activity Jogging	UCLA Questionnaire on jogging routines	Activity (1-10) Participation (%) Frequency (sessions/wk) Distance (km/session) Duration (min/session) Velocity (km/hour)	Joggers: 10±0; non-joggers: 6.6±2.4 a) Preop: 5.4%; postop: 3.8% 4 (1-7) 3.6 (0.5-15) 29 (5-90) 7.7 (3-18)
Alvarez et al. 2015	Spain	THA	Cross-sectional study	Patients: 47 Age: 63.8±11.7 55% male BMI: 29.6±5.6 19% bilateral	>1 year	Activity	UCLA IPAQ Accelerometer	Activity (1-10) IPAQ score PA (min/wk)	5.5±1.3 1.8±0.6 148.9±69.8
Amstutz et al. 2019	United States	MoM HRA	Survey	Patients: 661 Age: 51.9 (14-78) 70% male BMI: 26.5 (16.7-46.5)	10.1 (1-16) years	Sports participation	Questionnaire	Sports participation Type Activity quantity (frequency*duration)	100% Walking/hiking, cycling, gym workouts most-performed 12.4 (0-48) - Frequency (1: 1-4x/mo; 2: 5-8x/mo; 3: 9-12x/mo; 4: >12x/mo) - Duration (1: 0-30min; 2: 30-60min; 3: 60-120min; 4: >120min)
Banerjee et al. 2010	Germany	HRA	Case series	Patients: 138 Age: 52.6 (38-71) Male hips: 59% 10% bilateral	23.5 (12-42) months	Activity Sports participation	Grimby scale Questionnaire	Activity (1-6) Sports participation rate # sport activities	Preop: 3.3, postop: 4.6, p< .001 Preop: 98%, postop: 97% Preop 3.6 (0-10), postop: 3.2 (0-8)
								Duration of activities (<2, 2-4, >4 hours) Impact of activities (low=1, medium=2, high=3) Sports type	65% same, 25% increased, 10% decreased duration Preop: 2.1, postop: 1.5, p<0.001 Impact: 58% performed sports in same impact category, 5% in higher category, 37% in lower category Several high-impact activities showed decreased participation rate, several low-impact activities had higher participation rates Longer participation in less vigorous activities
Batailler et al. 2019	France	THA	Comparative retrospective study	Patients: 32 Age: 60.7±9.6 66% male BMI: 26±4 100% bilateral	20.1±11.6 months	Activity Sports participation	UCLA Questionnaire	Activity (1-10) RTS (%) Time to RTS, months RTS intensity	Preop: 6.1±1.7; postop: 5.6±8.1 87.5% returned to sports 4.2±2.7 68% same intensity
				Patients: 64 Age: 61.8±10 66% male BMI: 25.9±4 0% bilateral	28.9±15 months	Activity Sports participation	UCLA Questionnaire	Activity (1-10) RTS (%) Time to RTS, months RTS intensity	Preop: 5.7±1.9; postop: 4.9±2.2 57.8% returned to sports 5.4±5.1 months 56.5% same intensity
Bercovy et al. 2015	France	TKA	Therapeutic study	Patients: 403 Age: 70.36 (40.11-91.2) 34% male BMI: 29.6 (19.8-47.6) 23% bilateral	7.5 (5-13) years	Activity Sports activity	UCLA Questionnaire	Activity (1-10) # sports, n	Pre-symptom: 7.4±1.4 Preop: 3.8 Postop: 7.3±1.7 (1-10) 80% ≥7; 27% ≥8 72% ≥1 sports

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Bin Sheeha et al. 2020	Saudi Arabia / United Kingdom	TKA	Prospective cohort study	Patients: 33 Male: 6 Age: 76±7 (63-85) BMI: 32.38±2.01 Female: 27 Age: 59±6 (49-76) BMI: 37.21±7.65	12 months	Activity	Accelerometer	Stepping time (hour) Steps (steps/day) Time sedentary (hours) Time standing (hours) Upright time (hours) Cadence: <60 steps/min (min/wk) Slow to medium (60-100 steps/min) (min/wk) Moderate/vigorous (>100 steps/min) (min/wk) Meeting guidelines 150 min MVPA: All stepping events, (%) Walk events >10min (%) Minimal 5 min bouts (%) Minimal 1 min bouts (%)	Preop: 0.98±0.43; postop: 1.36±0.38 Preop: 4240±2268; postop: 6174±2287 Preop: 19.48±1.51; postop: 19.08±1.54 Preop: 3.47±1.27; postop: 3.54±0.97 Preop: 4.48±1.45; postop: 4.88±1.47 Preop: 369±189; postop: 593±188 Preop: 232±148; postop: 457±179 Preop: 6.6; Postop: 41.7 Postop: 24% Postop: 6% Postop: 9% Postop: 12%
Blikman et al. 2013	The Netherlands	THA (43%) TKA (45%) THA & TKA (11%)	Psychometric assessment	Patients: 44 Age: 72±9 23% male BMI: 28±5.4	>1 year	Activity	IPAQ short-form IPAQ long-form Accelerometer	Total activity (MET min/wk) Total time (min/wk) MPA (min/wk) VPA (min/wk) Sitting time (min/wk) Total activity (MET min/wk) Total time (min/wk) MPA (min/wk) VPA (min/wk) Sitting time (min/wk) Mean counts/min Total activity (min/wk) MPA (min/wk) VPA (min/wk) Sitting time (min/wk)	3370±3762 868±969 822±945 46±129 3123±1214 4498±3840 1108±922 1087±909 20±59 3255±1118 210±142 99±156 92±127 7±35 1252±314
Bonnin et al. 2010	France	TKA	Case series	Patients: 347 Age at FU: 74.8 (28-94) 35% male BMI: 27.9±4.6 0% bilateral	44 (13-71) months	Sports participation	Questionnaire	Sports activity (%) Sports impact (low, medium, high) in patients aged <75 (n=141) Sport type	41.5% increased activity level, 26.8% decreased 56% participated regularly in light, 66% in intermediate, 10% in strenuous sport You do not list this Cycling, hiking, swimming most performed
Bonnin et al. 2018	France	CoC THA	Case-control study	Patients: 1042 Age: 60.6±8.8 57% male BMI: 26.6±4.6 16% bilateral	44.4±14.0 months	Sports participation	Questionnaire	Sports participation (%) Sports impact: light, moderate, strenuous (%) Discomfort (none, moderate, severe) Sport level Sport type	91% 51% participated in ≥1 light sport, 71% in moderate, 20% in strenuous sport. 1-5% in light sport, 5-8% in moderate sport, 8-18% in strenuous sport reported severe discomfort during activity 2.6% played at competitive level Hiking, cycling, swimming most performed

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Bonnin et al. 2020	France	THA	Comparative study	Patients: 259 with posterolateral approach Age: 61±7.8 (39-78) 63% male BMI: 27±4.2 (17-41)	44±14.1 (24-74) months	Sports participation	Questionnaire	Participation (%) Sport impact: light, moderate, strenuous (%) Sport type	81% participated regularly/frequently 49% participated in ≥1 light sport, 69% in moderate sport and 19% in strenuous sport. Hiking most-performed (41%), followed by cycling
				Patients: 259 with anterolateral approach Age: 62±8.1 (21-76) 59% male BMI: 27±4.6 (18-45)	43±14.5 (24-79) months	Sports participation	Questionnaire	Participation (%) Sport impact: light, moderate, strenuous (%) Sport type	81% participated regularly/frequently 49% participated in ≥1 light sport, 71% in moderate, 24% in strenuous sport. Hiking most-performed (39%), followed by cycling
Brandes et al. 2011	Germany	TKA	Longitudinal follow-up study	Patients: 53 Age: 65.8±5.8 34% male BMI: 30.7±4.1 0% bilateral	1 year	Activity	Activities of daily living monitor Step monitor	Time walking %, n=32 Gait cycles/day, n=44 Gait intensity	Preop: 8.4±3.6, postop: 12.0±4.7 Preop: 4993±3170 gait cycles/day Postop: 5932±2111 gait cycles/day Walking happened at a higher intensity postop
Breuer et al. 2020	Austria	Short-stem THA	Prospective cohort study	Patients: 55 Age 61±10 53% male BMI: 27.9±4.9	38±4.6 months	Activity Sports participation	UCLA	Activity (1-10) Sports participation (%) Number of sports (n) Sports impact (low, high, mixed), %	Preop: 4.5±1.8; postop: 6.9±1.9 98% after mean of 13±8 weeks. 89% returned, 9% started sports activity Preop: 1.89±1.1; postop: 1.7±1.1 Preop: 28%, 43%, 17%; postop: 47%, 34%, 11%
Canetti et al. 2018	France	UKA	Comparative retrospective study	Patients: n=11, UCLA ≥5 Age: 66.5±6.8 BMI: 24.2±4.3 18% male Surgery: Robotic UKA	34.4±10.5 months	Activity Sports participation	UCLA Face-to-face questioning	Activity (1-10) RTS, % Time to RTS, months Number of sports Intensity level Sport type	6.4±1.6 100% 4.2±1.8 months 2.4 different sports 91% no change Hiking, cycling and fitness most performed
				Patients: n=17, UCLA ≥5 Age: 59.5±9.9 BMI: 26.3±3.8 28% male Surgery: normal UKA	39.3±15.5 months	Activity Sports participation	UCLA Face-to-face questioning	Activity (1-10) RTS, % Time to RTS, months Number of sports Intensity level Sport type	5.8±0.9 94% 10.5±6.7 months 1.9 different sports 82.4% no change Hiking, cycling, fitness most-performed
Casazza et al. 2020	United States	TKA	Prospective study	Patients: 7 Age: 55.6, SEM 3.5 39% male BMI: 32.8, SEM 1.5	12 months	Activity	Accelerometer	Steps (steps/day) Energy expenditure (kcal/day)	Preop: 5858 SEM 671; postop: 5981 SEM 751 Preop: 2377 SEM 229; postop: 2297 SEM 133
Chang et al. 2014 & 2015	South-Korea	TKA	Case series	Patients: 369 Age: 68.8 (50-83) 8% male BMI: 27.4 (19.3-39.1) 68% bilateral	2 (1-3) years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Sports participation (%) Number of activities (n) Sport impact Sport type	4.8±1.4 24% no regular sport activities, UCLA 4.0±1.9 76% no regular sport activities, UCLA 4.6±1.7 1.7 activities Decrease in high-impact sports, increase in moderate-impact sports Walking, swimming, cycling most-performed

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Clement et al. 2019	United Kingdom	THA	Comparative study	Patients: 200 Age: 69.9±9.2 (42-92) 43% male	12 months, 24 months	Activity	LEAS Accelerometer	Activity (1-18) Energy expenditure (MET/day) Time spent sitting/lying (hrs/day) Time spent standing (hrs/day) Time spent walking (hrs/day) Step count (steps/day)	*Patients aged <65* Preop: 8.7±2.4; 12mo: 12.5±2.8; 24 mo: 12.0±3.1 32.8±1.3; 12 mo: 31.1±1.4; 24 mo: 30.4±1.2 Preop: 18.9±1.8; 12 mo: 18.3±2.2; 24 mo: 18.6±2.0 Preop: 3.9±1.5; 12 mo: 3.4±1.6; 24 mo: 3.0±1.3 Preop: 1.2±0.6; 12 mo: 1.2±0.5; 24 mo: 1.0±0.6 Preop: 5583±3211; 12 mo: 5370±2524; 24 mo: 4269±2905
								Activity (1-18) Energy expenditure (MET/day) Time spent sitting/lying (hrs/day) Time spent standing (hrs/day) Time spent walking (hrs/day) Step count (steps/day)	*Patients aged 65-74* Preop: 7.9±2.6; 12 mo: 11.5±2.9; 24 mo: 11.6±3.2 Preop: 32.3±1.4; 12 mo: 32.2±1.7; 24 mo: 31.1±1.5 Preop: 18.4±2.7; 12 mo: 17.6±2.9; 24 mo: 17.6±2.7 Preop: 4.5±2.3; 12 mo: 4.4±2.0; 24 mo: 4.3±2.2 Preop: 1.0±0.7; 12 mo: 1.3±0.7; 24 mo: 1.2±0.6 Preop: 4501±3273; 12 mo: 6071±3256; 24 mo: 5760±3199
								Activity (1-18) Energy expenditure (MET/day) Time spent sitting/lying (hrs/day) Time spent standing (hrs/day) Time spent walking (hrs/day) Step count (steps/day)	*Patients aged ≥75* Preop: 7.5±2.2; 12 mo: 8.7±2.9; 24 mo: 9.0±3.0 Preop: 31.7±0.9; 12 mo: 31.4±1.4; 24 mo: 29.9±1.5 Preop: 19.3±3.1; 12 mo: 18.5±1.3; 24 mo: 18.1±1.3 Preop: 4.0±3.1; 12 mo: 3.8±1.0; 24 mo: 3.3±0.8 Preop: 0.6±0.4; 12 mo: 0.8±0.4; 24 mo: 0.9±0.4 Preop: 2740±1977; 12 mo: 4344±2195; 24 mo: 3915±2054
Cowie et al. 2013	United Kingdom	THA (90%) HRA (10%)	Case series	Patients: 239 Age: 55.2±7.2 35.2% male BMI: 28.2±4.9 19% bilateral	3.1±0.97 years	Activity Sports participation	Grimby scale Questionnaire	Activity (1-6) RTS time (weeks) Activity type	3.75±0.86 46% light exercise, 31.2% moderate <2hrs/wk, 6% moderate or hard exercise >2 hrs/wk 18.8±8.5 (Exercise) walking, swimming, gym most commonly reported
Crawford et al. 2020	United States	TKA	Retrospective review	Patients: 1611 Age: 63.9 34% male BMI: 33.9 27% bilateral	11.4 (5.1-15.9) years	Activity	UCLA	Activity (1-10)	7.74±1.70
Daugaard et al. 2018	The Netherlands (TKA data)	TKA	Cross-sectional study	Patients: 52 Age: 66±7.2 50% male BMI: 29.5±4.6	5 years	Activity	Accelerometer	Number of steps Number of short walking bouts (<10sec) Number of transfers (sit/stand)	6702±3203 steps/day, n.s. compared to healthy adults 48.7±25.9, significantly less than healthy adults 43±16.8, n.s. compared to healthy adults
Delfin et al. 2017	Sweden	THA	Matched control study	Patients: 27 Age: 72.6±7.1 37% male BMI: 27.2±4.1	12.6±5.3 years	Activity	UCLA Questionnaire	Activity (1-10), median IQR Intensity (1: no PA, 5: VPA >1x/wk)	6 [4-7] 4 [3-4]

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Donner et al. 2019	Germany	Bilateral short-stem THA	Case series	Patients: 51 Age: 63.1 (36.7-76.8) 56.9% male BMI (median): 27.6 (16.6-41.8) 100% bilateral	4.9 (4.3-5.8) years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Participation rate (%) Activity (hours) Sport activity (hours) Frequency (n) Number of activities (n) Activity type	Preop: 3.8±2.0, postop: 4.7±1.7, p= .001 Preop 60.8% did sports, postop 76.5% Preop: 3.4, postop: 4.2, p=.26 Preop: 3.4±2.4, postop: 3.7±4.1, p=.44 Preop: 2.8±3.1, postop: 3.4±2.9, p=.21 Preop: 2.3, postop: 1.8. Postop more participation in low-impact, less in high-impact sports compared to preop
Dubin et al. 2020	United States	TKA	Case series	Patients: 121, LEAS <10 preop Age: 61.7 43% male BMI: 33.2	2.0 years	Activity	LEAS	Activity (1-18)	Preop: 7.1 Postop: 8.7
				Patients: 127, LEAS ≥10 preop Age: 61.7 49% male BMI: 32.2	2.0 years	Activity	LEAS	Activity (1-18)	Preop: 11.8 Postop: 10.8
Eckhard et al. 2020	Australia	TKA	Prospective study	Patients: 563 preop Age: 68.0±8.5 50% male BMI: 31.3±6.6 Patients: 381 postop Age: 68.0±8.0 47% male BMI: 31.2±6.3	1 year	Activity	UCLA	Activity (1-10)	Preop: 4.3±1.7 Postop: 5.9±1.7
Elman et al. 2014	Dominican Republic	THA (22%) TKA (81%)	Case series	Patients: 64 Age: 61.1±13.9 20% male BMI <35 kg/m2: 23% 41% bilateral	2.1 (1-4) years	Activity	YPAS	Mean total time summary index (hours/wk) Mean energy expenditure summary index (kcal/wk) Participate in PA (min/wk) MVPA (min/wk)	23.6±29.1 5178.9±6552.4 189.6±233.1 84.6±158.9
Felts et al. 2010	France	Medial UKA	Therapeutic study	Patients: 62 Age: 54.7±5 47% male BMI: 28±4 5% bilateral	11.2±5 years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Activity type Sport level RTS (%)	6.4±2.3, 40% ≥8 Hiking, cycling, swimming most-performed All patients except two returned to their presymptomatic activity level 83.4%
Fisher et al. 2011	United Kingdom	HRA	Case series	Patients: 117 Age: 54 (30-73) 57% male 14% bilateral	30 (16-50) months	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Participation rate RTS (%) Sport type	Preop: 4.4 (1-10); postop 6.8 (3-10), p<.01 73% 87% of 86 (73%) patients active preop returned to sports, 32% picked up sport Swimming, cycling, golf most popular preop & postop
Foucher et al. 2018	United States	THA	Cross-sectional cohort study	Patients: 16 Age: 56.8±8.3 31% male BMI: 30.9±7.7	23.8 ±10.2 (12-48) months	Activity	UCLA	Activity (1-10)	6±2 (mode: 3) (most common response: 3, 2 out ot 10 subjects (12.5%)

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Fouilleron et al. 2012	France	HRA	Case series	Patients: 40 runners preop Age: 50.7 (31-61) 90% male BMI: 24.8 (21.7-33.6) 8% bilateral	33.3 (26-41) months	Running activity	UCLA Devane activity score Questionnaire	Activity (1-10) Activity (1-5) Return to running (%) Number of sports (n) Time jogging (hrs/wk) Distance (km/wk)	Preop: 6.9 (5-10), postop: 9.1 (4-10), p<.001 Preop: 3.7 (3-5) postop: 4.6 (2-5), p<.001 91.6%, mean 16.4 weeks postop Preop: 4.2, postop: 3.8, p=.196 3.1 Preop: 38.9, postop: 28.4
Fujita et al. 2013	Japan	THA	Case-control series	Patients: 38 Age: 60.9±9.1 0% male BMI: 23.0±3.6 13.2% bilateral	12 months	Activity	Pedometer	Steps/day, % healthy controls LPA (min/day), % healthy controls MPA (min/day), % healthy controls VPA (min/day), % healthy controls	Preop: 4632±2246, 65%; 79%; 12mo: 6163±2410, 85% Preop: 107±49, 78%; 12mo: 125±42, 90% Preop: 16±18, 27%; 12mo: 46±50, 77% Preop: 1±3, 8%; 12mo: 3±4, 25%
Gerhardt et al. 2017	The Netherlands	HRA	Prospective cohort study	Patients: 38 Age: 54.4±9.4 55% male BMI: 26.1±3.1	1, 2, 3 years	Activity	UCLA	UCLA (1-10)	Preop: 5.0 (4.0-7.0) 1yr: 7.5 [6.5-9.0] 2yr: 7.0 [7.0-8.5] 3yr: 7.0 [6.5-8.0]
Ghomrawi et al. 2017	United States	THA (52%) TKA (48%)	Retrospective cohort study	THA: 403 patients (85 had 2yr FU) Age: 65.5±10.8 41% male TKA: 364 patients (69 had 2yr FU) Age: 67, 1±9.4 33% male	2 years	Activity	UCLA LEAS	Activity (1-10) Activity (1-18)	THA: Preop: 4.4±2.1, postop: 6.0±2.0 TKA: Preop: 4.5±2.1; postop: 5.9±2.1 THA: Preop: 9.4±3.1, postop: 11.6±3.3 TKA: Preop: 9.1±2.8, postop: 10.9±3.0
Girard et al. 2013	France	HRA	Prospective case series	Patients: 50 did high-impact sports preop Age: 51.5 (30.8-64.8) 90% male BMI: 23.7 (21.7-33.6) 10% bilateral	44.1 (39.1-54.5) months	Activity Sports participation	UCLA Questionnaire	Activity RTS (%) RTS time (wk) Sport type Number of activities (n) Time spent (<2, 2-4, 4-7, >7), (hrs/wk)	Preop: 6.6 (4-10), postop: 9.1 (8-10), p< .001 98% (82% high-impact) 14,6 (7-29) Jogging, soccer most-performed of the high-impact sports Preop: 4 (1-7), postop: 3.6 (1-8) Preop: 3.2 (1-8), postop: 2.9 (2-9)
Groen et al. 2012	The Netherlands	TKA	Observational study	Patients: 830 Age: 72±9 25% male 0% bilateral	(1-6) years	Activity	SQUASH	Activity (min/wk) Meeting health recommendation (%) Meeting fitness recommendation (%) Meeting health & fitness recommendation (%)	1337±1260 51% 53% 46%
Hara et al. 2018	Japan	THA	Cohort study	Patients: 524 Age: 62.9±10.1 16% male BMI: 22.9±3.3 19% bilateral	68.2 months	Activity Sports participation	UCLA Questionnaire	Activity Sports participation and impact Sport type	Preop: 3.8±2.1; postop: 4.6±1.7, p<.001 Preop: 15.5%, postop: 30.5%, mostly in low-impact sport, 2.5% high impact Most-reported: walking, swimming, gymnastics

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Hayashi et al. 2012	Japan	THA G2 stem	Prospective cohort study	Patients: 43 Age: 66.7±8.2 21% male BMI: 23.7±3.1 16% bilateral	12 months, 24 months	Activity	UCLA	Activity (0-10)	1yr: 5.4±2.1 2yr: 6.3±2.2
		THA VerSys FMT stem	Prospective cohort study	Patients: 15 Age: 57.8±6.8 BMI: 22.5±2.2 20% male 0% biltateral	12 months, 24 months	Activity	UCLA	Activity (0-10)	1yr: 6.3±1.5 2yr: 6.3±1.5
Hayashi et al. 2016	Japan	THA	Prospective cohort study	Patients: 65 Age: 65.1±10.4 17% male BMI: 23.4±3.5	24 months	Activity	UCLA	Activity (0-10)	5.9±1.6
Hayes et al. 2011	Australia	TKA	Cross-sectional study	Patients: 16 Age: 64.4±4.6 50% male BMI: 69.5±2.6	12 months	Activity	Intelligent device for energy expenditure and activity (IDEEA)	Energy expenditure (j/min/kg) Activity (% recording interval)	103 4.6%
Heiberg et al. 2016	Norway	THA	Long-term FU of RCT (controls)	Patients: 30 Age: 70.6±8.4 56.7% male	5 years	Activity Sports participation	UCLA Self-reported	Activity (1-10) Training sessions (n/wk)	7±1.9 3±1.0
Hepperger et al. 2018	Austria	TKA	Prospective single-cohort study	Patients: 200 Age: 72±7.7 40% male BMI: 27.2±5.0 17.5% bilateral	1 year 2 years	Activity Sports participation	Tegner activity scale Questionnaire	Activity, median [25%, 75%], (1-10) Frequency (none, occasionally, 2-3x/wk, >5x/wk) Sport type Sport impact (highest):	12mo: 3.0 [3.0, 4.0]; 24mo: 3.0 [3.0, 4.0] 12mo: 18% none, 13% occasionally, 59% 2-3x/wk, 10% >5x/wk 24mo: 17% none, 13% occasionally, 57% 2-3x/wk, 13% >5x/wk) Summer: swimming, hiking, cycling Winter: Hiking, skiing, cross-country skiing Summer: low: 9%; medium: 86%, high: 5% Winter: low: 3%; medium: 43%, high: 54%
Hjorth et al. 2018	Denmark	MoM THA MoM HRA MoP THA	Prospective cohort study	MoM THA/HRA: 77 Age at FU: 59.1 (51.3-64.4) 66% male MoP THA: 71 Age at FU: 68.3 (60.9-69.9)	MoM THA/HRA baseline: 7.1 (4.6-7.6) years MoP THA baseline: 8.9 (4.7-10.7) years? And baseline + 3, 6, 9, 12 months	Activity	Accelerometer	Mean time spent on walking, bicycling or high-impact activities during daily wear time (%)	MOM THA/HRA: BL + 3m: 12.66% [11.62-13.69] BL + 6m: 11.82% [10.81-12.83] BL + 9m: 11.12% [10.19-12.05] BL+ 12m: 13.15% [12.20-14.09] MoP THA:BL + 3m: 13.43% [12.13-14.72] BL + 6m: 12.33% [10.50-14.17] BL + 9m: 10.94% [9.35-12.53] BL+ 12m: 13.34% [11.66-15.03]

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Ho et al. 2016	United States	TKA UKA	Retrospective cohort study	TKA group Patients: 39 Age: 59 (57-64) 30% male BMI: 32.5 [28.8-38.4] 3% bilateral	4.0±1.2 years	Activity Sports participation	UCLA Phone interview	Activity (1-10) Participation (%) RTS (%), time to RTS (mo), new sport (%) Frequency Duration Sport type	6.1±1.4 70% 83%, 4 [2-12], 8% High range, most 4x or 12x/wk 30 min most common Cycling, dancing, swimming most-performed
				UKA group: Patients: 33 Age: 60 (53-64) 33% male BMI: 30.3 [27.6-33.7] 9% bilateral	4.0±1.2 years	Activity Sports participation	UCLA Phone interview	Activity (1-10) Participation (%) RTS (%), time to RTS (mo), new sport (%) Frequency Duration Sport type	7.4±1.6 72%, 87%, 2 [1.5-4.9], 0% 4x/wk most common 60 min/session most common Golf, cycling, hiking most-performed
Hodges et al. 2018	Australia	TKA	Secondary analysis of cohort study	Patients: 434 Age (at FU): 66.0±6.3 47% male	12 months	Activity	Active Australia Survey	Meeting activity guidelines ≥6 hrs/day sedentary Inadequate PA and sedentariness ≥6 hrs/day	47% 45% 24%
Hylkema et al. 2020	The Netherlands	TKA	Observational repeated-measures study	Patients: 57 employees working ≥24 hrs/wk Age: 59±4 (48-65) 47% male BMI: 20% normal, 33% overweight, 47% obese	1 year	Activity	Accelerometer	Sedentary (% wear time) Prolonged sedentary bouts, ≥30min (% wear time) LPA (% wear time) MVPA (% wear time) Meeting guidelines, 150 min/wk of MVPA (%)	60.1% 16.5% 36.8% 3.1% 70%
Ibrahim et al. 2019	Sweden	THA	Prospective cohort study	Patients: 42 Age female: 68.5 (54-85) Age male: 68 (50-80) 48% male	5 years	Activity	Tegner activity scale	Activity (1-10), median and range	Female: Preop: 1 (0-3), postop: 2 (1-4) Male: Preop: 1 (0-4); postop: 2 (1-5)
Innmann et al. 2016	Germany	THA	Retrospective cohort study	Patients: 86 Age: 52 (21-60) 61% male BMI: 27 (18-39) 4% bilateral	11 (10-12) years	Activity Sports participation	UCLA Schultness Clinic sports and activity questionnaire	Activity (1-10) Intense activity (≥7), (%) Sports participation (%) Disciplines (n) Frequency (times/wk) Duration (min) Sport type	Preop: 3.8±1.6; postop: 6.2±1.5, p< .001 41% 79%, 11% exercising preop stopped sports, 20% non-sporters preop started exercising Preop: 1.8; postop: 1.7 Preop: 2.3; postop: 2.6 Preop: 53, postop: 55 Walking most-performed, most other low-impact sports follow with 10-20% participation rate
Issa et al. 2015	United States	TKA	Prospective case series	Patients: 281 Age: 66 (39-80) 38% male 2% bilateral	1, 2, 3, 4, 5 years	Activity	Lower extremity activity scale	Activity (1-18)	Preop: 9.18, 1yr: 11.19, 2yr: 11.49; 3yr: 11.19; 4yr: 11.11; 5yr: 11.47
Jacquet et al. 2020	France	UKA	Retrospective case-control study	Patients: 50 did high-impact sports preop Age: 50.8±4.4	3.9±1.8 (2-5.7) years	Sports participation	UCLA	UCLA RTS time (months) Return to impact activities (UCLA >8) (%)	6.5±2 5.8±4.2 28% High tibial osteotomy superior to UKA considering RTS, sport impact and UCLA score.

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Jahnke et al. 2015	Germany	UKA	Case series	Patients: 135 Age: 67.1 (38-88) 53.4% male 9% bilateral	2±1.47 years	Activity Sports participation	UCLA (1-10) Tegner (1-10) Heidelberg Sports Activity Score (HAS)	Activity (1-10) Activity (1-10) HAS Sports participation Sport type	6.33±1.08, small increase 3.96±0.96, no change 29.29±18.48, high increase 93.2%, 84% 1x/wk or more Hiking, cycling, swimming most-performed. High-impact sports participation decreased postop.
Jassim et al. 2019	United Kingdom	THR TKR	Case series	THA: Patients: 40 Age: 53.1±8.4 47% male TKA: Patients: 24 Age: 60±2.5 42% male	THA: 3.3±1.1 years TKA: 3±0.9 years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) RTS Sport type Frequency	14% improved, 11% decreased UCLA. THA patients improved (preop: 7.78; postop: 7.93), TKA patients decreased (preop: 7.5; postop 7.29) activity. 100% Golf, running, gym most-performed postop. Remarkable increase in tennis, remarkable decrease in cycling Most participated in sports 4-6x/wk postop .
Jeldi et al. 2017	United Kingdom	THA	Observational cohort study	Patients: 27 Age: 67 (50-82) 33% male BMI: 31 (19-43)	1 year	Activity	Activity monitor	Steps/day	6155±2631
Jelsma et al. 2019	The Netherlands	MoM THA	Cohort study	Patients: 62 Age: 60.8±9.3 58% male BMI: 27.6 (20.2-37.9)	6.3 (3.7–9.6) years	Activity	Accelerometer	Time walking (s/day) Time cycling (s/day) Steps (n/day) High intensity (# peaks)	5403±2472 (1646–17126) 282±531 (0–2040) 6694±3011 (1712–15443) 78.4±197.5 (0–1406)
Jelsma et al. 2020a & Jelsma et al. 2020b	The Netherlands	THA HRA	Observational matched cohort study	Patients: 16 HRA Age: 55.5±9.7 (43-67) 75% male BMI: 26.1±3.8 (22-37) 0% bilateral	10 (9-11) years	Activity	SQUASH Accelerometer	Total activity score (MET/wk), median Time standing (hrs), median Time sitting (hrs), median Time walking (hrs), median Time cycling (hrs), median Steps (n/day), median Sit-stand transfers (n), median Time active %	6150 (1110-18480) 3.0 (1.8-5.7) 7.6 (4.6-12) 1.3 (0.5-1.9) 0.05 (0-0.48) 5546 (2274-9966), 31% < 5000, 13% 8000-10000, 6% >10000 39 (21-74) 10.8±7.0
				Patients: 16 THA Age: 60 (53-68) 75% male BMI: 29 (20-40) 0% bilateral	10 (8-12) years	Activity	SQUASH Accelerometer	Total activity score (MET/wk), median Time standing (hrs), median Time sitting (hrs), median Time walking (hrs), median Time cycling (hrs), median Steps (n), median Sit-stand transfers (n), median	4560 (1050-9300) 3.0 (1.6-6.2) 9.6 (3.8-13) 1.1 (0.4-0.8) 0.01 (0.0-1.2) 4600 (1567-11749), 50% <5000, 19% ≥8000 37 (24-65)
Jonas et al. 2019	United Kingdom	HRA	Matched control study	Patients: 51 Age 49.8 (18-67) 78% male BMI 25.7 (19.7-35.1)	17.6±0.53 years	Activity	UCLA	Activity (1-10)	8 [6-10]
		Hybrid THA	Matched control study	Patients: 53 Age 50.4 (21-66) 75% male BMI 27.0 (18.5-37.0)	19.9±0.62 years	Activity	UCLA	Activity (1-10)	6 [5-7]

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Jones et al. 2012	United States	TKA	Prospective cohort study	Patients: 83 Age: 66±9.7 46% male 0% bilateral	1 year	Activity Sports participation	Historical Leisure Activity Questionnaire (HLAQ) HLAQ	Activity (MET hrs/wk) Sport type Sport intensity	21.4±24.9 (0-125.6) MET hrs/wk: significantly more than preop but significantly less than predicted by patients preop. Walking, calisthenics, gardening, strength-training most-reported. Almost all activities performed at moderate intensity (3-5.9 MET), 19.6 of 21.4 MET hrs/wk 1.5±6.5 MET at high-intensity 0.1±0.9 (0-7.5) at low-intensity
Karampinas et al. 2017	Greece	SMF (38%) BFH (33%) HRA (31%)	Retrospective comparative non-blinded study	Patients: 48 Age: 65 (52-70) 60% male 15% bilateral	2 years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Impact level of sports (low, medium, high)	BFH: 6.7; HRA: 8.1; SMF: 7.9. All improved compared to preop. Most patients in all groups participated in low-impact activities.
Keeney et al. 2014	United States	TKA	Therapeutic study	Young patients (≤55): 181 Age: 49.0 26% male BMI: 34.4 21% bilateral Old patients (65-75): 262 Age: 69.9 40% male BMI: 31.5 20% bilateral	Young: 36 (12-118) months Old: 31 (12-110) months	Activity	UCLA	Sedentary (1-2), mildly active (2-4), moderately active (5-7), highly active (8-10)	Young: Mean UCLA score was 4.6, 57% improved activity level, 14% decreased activity level. 56% sedentary or mildly active, 10% returning to regular activity Old: Mean UCLA score was 4.9, 65% improved activity level, 14% decreased activity level. 52% sedentary or mildly active, 12% returning to regular activity
Keeney et al. 2015	United States	THA	Therapeutic study	Young patients (≤50): 704 Age: 39.0 51.1% male BMI: 29.1±6.8 17% bilateral Old patients (65-75): 484 Age: 69.4 40.3% male BMI: 29.0±5.5 7% bilateral	Young: 52 (12-136) months Old: 35 (12-156) months	Activity	UCLA	Sedentary (1-2), mildly active (2-4), moderately active (5-7), highly active (8-10)	Young: Mean UCLA score was 6.4±2.2. 81% did at least in moderate activities, 37% did impact activities. Old: Mean UCLA score was 5.3±1.9, 61% did at least moderate activities, 15.5% did impact activities.
Kersten et al. 2012	The Netherlands	TKA	Cohort study	Patients: 830 Age: 72.0±9.3 25.9% male BMI: 29.4±5.0	3.0±1.2 years	Activity	SQUASH	Total activity (min/wk) LPA (min/wk) MPA (min/wk) HPA (min/wk) Household (min/wk) Sport activities (min/wk) Walking (min/wk) Cycling (min/wk) Meeting guidelines (%)	1347±1278 780±874 337±577 223±374 715±794 52±140 167±335 122±242 50.8

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Kim et al. 2016	South Korea	Short-stem THA	Case series	Patients: 400 Age: 53±13 66% male BMI: 29±4 33% bilateral	Short stem: 17.8 (13-20) years	Activity	UCLA	Activity (1-10)	8.6 (8-10)
		Ultra-short stem THA	Case series	Patients: 201 Age: 53±9 59% male BMI: 30±4 10% bilateral	Ultra-short stem: 12.3 (10-13) years	Activity	UCLA	Activity (1-10)	9 (8-10)
Kim et al. 2019	South Korea	UKA	Prospective comparative study	Patients: 42 Age: 63.6±5.5 17% male BMI: 25.3±2.4 0% bilateral	24 months	Activity Sports participation	UCLA Tegner Questionnaire	Activity (1-10) Activity (1-10) Sports participation % Time to return (months) Number of sport Sessions (n/wk) Session duration Level of impact	~5.1 ~3.6 76.2% 3.8 months postop 2.2±0.8 3.0±1.3 1.3±0.7 Decrease in high-impact activities, increase in swimming & gymnastic riding
Kiran et al. 2019	United Kingdom	HRA	Prospective case series	Patients: 66 Age: 45.7 (36.8-62.8) 62% male BMI: 27.65±4.3 9% bilateral	1 year, 10.6±0.5 years	Activity	UCLA	Activity (1-10)	1yr: 8.07±0.93 10yr: 7.07±1.16 All patients were able to return to their pre-arthritic level of activity, this included some high-level amateur sports.
Kleeblad et al. 2020	United States	UKA	Case series	Patients: 164 Age: 62.3±8.8 55% male BMI: 27.6±4.4 9% bilateral	20.2 months	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Participation (%) RTS (%) Level of sport Time to return (months) Number of sports (n) Sport type Impact level (% participants)	Preop: 5.93±2.19; postop: 6.78±1.92 89.6 % 95% 42% increased, 43% similar 4.43±3.54 3.6±3.0 Cycling (45%), swimming (38%) most-performed 32.7% performed high-impact sports, 63.9% medium-impact, 93.9% low-impact
Krantz et al. 2012	France	HRA	Case series	Patients: 22 Age: 24.9 (17.1-29.9) 31% male BMI: 24.2 (18.8-36.2) 9% bilateral	50.6 (44-59) months	Activity	UCLA	Activity (1-10)	Preop: 5.5 (1-9); postop: 7.6 (1-10)
Krych et al. 2017	United States	UKA	Therapeutic study	Patients: 183 Age: 49.2 55% male BMI: 32.4	1, 2, 5, 5.8 years	Activity	Tegner activity scale	Activity (1-10)	Preop (n=183): 2.65 1yr (n=76): 4.07 2yr (n=133): 4.33 5yr (n=67): 4.48 5.8yr (n=48): 4.48 UKA scored higher than proximal tibial osteotomy at 3 months, 2 years and final FU (5.8yr for UKA, 7.2yr for proximal tibial osteotomy)

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Kuhn et al. 2013	United States	THA	Prospective case series	Patients: 37 Age: 42.1±7.7 (17.8-50.3) 32% male BMI: 29.0±5.6 (20.1-44.3)	1.3±0.2 years	Activity	UCLA Stepwatch activity monitor	Activity (1-10) Daily strides Inactive time (%/day) Low-intensity (%/day) Medium-intensity (%/day) High-intensity (%/day)	Preop: 6.0±2.2, postop: 7.2±1.6 Preop: 4541±1594; postop: 5584±1626 Preop: 74.8%, postop: 71.1%±6.1, p<.001 Preop: 16.8%, postop: 18.7%±4.1, p=.004 Preop: 7.4%, postop: 8.8%±2.7, p=.001 Preop: 1.0%, postop: 1.4%±0.9, p=.006
Le Duff & Armstutz, 2011	United States	HRA	Prospective case series	Patients: 201 Age: 49.6 74.6% male BMI: 26.9 (19-46) 28% bilateral	1.8 (1.0-4.9) years 9.1 (5.0-13.4) years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Sports participation Number of activities Activity type Activity impact Activity level Frequency and duration	1.8yr: 7.8; 9.1yr: 7.7 1.8yr: 96%; 9.1yr: 90%, p= .50 1.8yr: 2.47; 9.1yr: 2.13, p= .02 Walking, cycling, weightlifting, swimming, golf most popular at both measurement times High-impact represented 12.4% after 1.8yr and 17.5% and 9.1yr, p=.045 Competitive playing in 5.5% after 1.8yr and 10.1% after 9.1yr; 33.9% after 1.8yr and 33.3% after 9.1yr participated in their favorite activity >12x/month, with 45% and 36% of sessions lasting 30 and 60 min, respectively.
Lefevre et al. 2013	France	THA TKA	Cross-sectional study	THA group: Patients: 27 Age: 63±7.2 33% bilateral TKA group: Patients: 8 Age: 72.8±5.2 25% bilateral	THA: 8.8±7.1 years TKA: 7±5.9 years	Judo	Mailed survey	Return to judo Time to return Intensity Frequency (times/wk)	THA: 81.5%; TKA: 62.5% THA: 3.9±2.7; TKA: 5.2±2.2 All participants stopped competitive judo 2.5±1
Lo Presti et al. 2019	Italy	Medial UKA	Prospective case series	Patients: 53 did ≥1 sport preop Age: 59.7 (46-66) 28.3% male	48±6 (24-72) months	Sports participation	Questionnaire	RTS (%) Sport type Impact (high/low) Frequency (times/wk) Session length (min) Intensity	90%, 13.2% within 3 months, 35.8% within 6 months Cycling (29%), swimming (31%) most-commonly reported. Preop. 32% participated in high-impact sport; postop. 21% Preop: 2.1; postop: 1.9 Preop: 50; postop: 43 None played professional sports, all sports performed as recreational activities
Long et al. (2014)	United States	TKA	Therapeutic study	Patients: 88 (36 filled out Tegner scale) Age: 51 (22-55) 25% bilateral	8 (3-18) years 25.1 (20-35) years	Activity	Tegner activity scale	Activity (1-10)	Preop: 1.5; 8yr: 3.5±1.1; 25yr: 3.0±1.6 Patients with no contralateral hip pain or knee pain had a 3.5 Tegner score at both time points
Lubbeke et al. 2014	Switzerland	THA (97%) HRA (3%)	Prospective single-cohort (BL to 5yr comparison) + cross-sectional study (10yr FU)	5yr FU (n=1085): Age: 67.1±11.7 44.6% male BMI: 27.0±4.6 10yr FU (n=757): Age: 65.1±11.0 45.8% male BMI: 26.7±4.2	5 years, 10 years	Activity	UCLA	Activity (1-10), UCLA ≥7 (%)	Preop: 3.5, 4.9%; 5yr: 5.7, 27.9%; 10yr: 5.5, 28.2%

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Lutzner et al. 2014	Germany	TKA	Prospective randomized study	Patients: 97 Age: 68.9 (CI: 67-70) 54% male BMI: 31.3 (CI: 30.3-32.3)	1 year	Activity	Accelerometer (activPAL activity monitor)	Steps/day MVPA (steps/day) Time spent lying (hrs/day) Time spent sitting and standing (hrs/day) Walking time (hrs/day) Meeting guidelines (%) Comparison with healthy population	Preop: 5278±2777; postop: 6473±3654 Preop: 1150±982; postop: 1935±1728, ~19 min/day Preop: 12.2±2.4; postop: 12.2±2.7 Preop: 10.8±3.7; postop: 10.3±2.6 Preop: 1.4±1.3; postop: 1.5±0.8 Postop: 16.5% In the healthy population more steps were taken in total and in MVPA
Lutzner et al. 2016	Germany	TKA	Case series	Patients: 221 Age: 68.1±9.5 43.4% male BMI: 31.3±4.9	1 year	Activity	Accelerometer (activPAL activity monitor)	Steps/day MVPA (steps/day) Time spent lying (hrs/day) Time spent sitting and standing (hrs/day) Walking time (hrs/day) Meeting guidelines (%) Lifestyle (<5000 sedentary, 5000-7500 low-active, >7500 active)	Preop: 5371±2820; postop: 6587±3299; Preop: 1149±1128; postop: 1835±1646 Preop: 12.3±2.3; postop: 12.0±2.4 Preop: 10.6±2.9; 10.4±2.3 Preop: 1.4±1.0; postop: 1.6±0.7 22.6% Preop: 48.0% sedentary, 34.8% low-active, 17.2% active; postop: 34.8% sedentary; 34.4% low-active, 30.8% active; p<.001
Madrid et al. 2019	Colombia	THA	Cross-sectional study	Patients: 535 Age: 67 (13-91) 31% male BMI: 25.5±3.9	>1 year	Sports participation (institutionalized)	Phone survey	Sports participation RTS (%), time (months) Sport type Sports activity (<5, 5-10, 10-20, >20), (hrs/wk) Level	Preop: 13.5%; postop: 7.1% 44%, 2.60±1.2 (1-4) Golf, hiking, swimming most-performed. Increase in hiking, decrease in jogging, soccer, tennis Preop: 39%, 35%, 25%, 3%; postop: 42%, 37%, 21%, 0% 71% perceived performance equal or better postop
Majewski et al. 2014	Switzerland	THA	Cohort study	Patients: 64 Age: 60 Male: 42 (65.6%)	>10 years FU Mean FU 148 months (120-191)	Sports activity	Sports activity index	Sports activity (0-6= non-active; 7-10= moderately active; 11-25= active)	Mean: 13 points Nonactive: 31% (13% scored 0/2, 11% scored 3/4, 7% 5/4) Moderately active: 38% (11% scored 7/8, 26% scored 9/10) Active: 31% (16% scored 11/12, 11% scored 13/14, 11% scored 15/16, 2% 17/18
Malcolm et al. 2014	United States	THA	Cross-sectional study	Young patients (≤30): 70 Age: 23.3 (12.8-30.7) 40% male 41% bilateral	6.6 (2.1-14.9) years	Activity	UCLA	Activity (1-10)	6.5±2.3
				Old patients (>30): 158 Age: 63.6 (33.6-91.5) 49% male 23% bilateral	5.3 (2.0-10.8) years	Activity	UCLA	Activity (1-10)	6.4±2.0
Martin et al. 2018	United Kingdom	HRA	Case series	Patients: 80 Age: 54±8.5 100% male	1 year	Activity	UCLA	Activity (1-10)	Preop: 5.4±2.1; postop: 7.2±1.7; p<.001
Matsunaga-Myoji et al. 2020a	Japan	TKA	Prospective case series	Patients: 58 Age: 72.6±6.0 16% male BMI: 26.1±4.4 58.6% bilateral	2 years	Activity	Accelerometer	Meeting PA guidelines for older persons (MVPA ≥52.5 min/wk) (%) Light PA (min/wk) MVPA (min/wk) Steps (steps/day)	41.4% Preop: 267 (159-368); postop: 330 (241-469) Preop: 8.2 (1.3-29.7); postop: 42.8 (10.0-89.2) Preop: 3542 (1958-4828); postop: 4687 (3147-7293)

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Matsunaga-Myoji et al. 2020b	Japan	THA	Prospective case series	Patients: 107 Age: 61.4±8.1 14% male BMI: 23.1±3.5 0% bilateral	1 year 3 years	Activity	Accelerometer	Steps (steps/day) MVPA (min/wk) Meeting guideline of 150min/wk MVPA (%)	Preop: 4776±2259; 1yr: 6634±3242; 3yr: 6736±3150 Preop: 35.4±35.1; 1yr: 58.3±64.6; 3 yr: 72.3±67.4 Preop: 2.0%, 3yr: 18.0%
Mayr et al. 2015	Germany	TKA	Retrospective case series	Patients: 81 Age at FU: 71.8±5.4 47% male BMI: 28.4	6.4±0.9 years	Sports participation	Questionnaire	Sport type Frequency (n/wk) Duration (hrs/wk) Sport impact	Biking, swimming, mountain hiking most performed sports Preop: 2; postop: 3.5 Preop: 3.3; postop: 5.3 25% participated in high-impact sports, 47% in medium-impact sports, 52% in low-impact sports
Meding et al. 2012	United States	TKA	Retrospective therapeutic study	Patients: 62 Age: 63.8±8.9 27% male 58% bilateral	21.1±1.6 years	Activity	UCLA	Activity (1-10)	8.3±1.2 (5-10) 35% UCLA 9 or 10
Mesko & Heath 2011	USA	CoC-THA	Cohort study	Patients: 62 Age: 52 (26-70) 62.5% male BMI: 29.5 (20-50) 7% bilateral	121 (98-168) months	Activity	UCLA	Activity (1-10) High-impact (UCLA 9/10), (very) active (UCLA 7/8), inactive/moderately active (UCLA 1-6)	6.6±2.04 16% performed high-impact sports, 35% highly active. 48% inactive to moderately active
Naylor et al. 2019	Australia	TKA THA	Prospective case series	Patients: 1289 Age: 67.2±9 45% male BMI: 30.9±6 5% bilateral	3 years	Activity	Phone interview	Participation in PA ≥1x/wk (%) Sport type	THA: Preop: 50.6; postop: 67.3, p< .001 TKA: pre: 40.8; postop: 60.6, p< .001 Walking performed most (47%). General increase in low-impact activities (walking, cycling, gym, swimming), general decrease in high-impact activities (tennis, squash, jogging)
Ninomiya et al. 2018	Japan	THA	Cross-sectional controlled study	N: 58 Age: 68.0±5.4 17.3% male BMI: 22.1±2.8 100% unilateral	10.4±0.4 years	Activity	IPAQ	Activity (high ≥1000kcal, low <1000kcal)	High: 34.4%, low: 65.5% Healthy control group: High: 69.6%, low: 30.4%
Ollivier et al. 2014	France	THA	Retrospective case series	Patients: 571 Age: 61.3±10.9 Gender: 52% male BMI: 27±3.2 0% bilateral	9.8±2.9 years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Sports participation RTS time (months) Sport type	6.8±2.2 (1-10) 64% 6.6±3.2 (2-48) Highest frequencies in low-impact sports, 26.6% participated in non-recommended activities
Ortmaier et al. 2017	Austria	Short-stem THA	Retrospective case series	Patients: 137 Age: 65.6±12.4 BMI: 26.6±4	20.4±2.3 (18-22) months	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Participation (%), RTS (%) Time to RTS Disciplines (n) Sport type Frequency (0, 1, 2, 3, 4, >4x/wk), (%) Session length (0-15, 15-30, 30-60, 60-120, >120 minutes), (%) Sport level	7.1 (4-10) Preop: 92%; postop: 87%, RTS: 91% 62% within 3mo, 86% within 6mo Preop: 2.9±2.1; postop: 2.6±1.9 Swimming, hiking, cycling, nordic walking, alpine skiing most-performed, decrease in most high-impact sports 18%, 12%, 20%, 14%, 9%, 27%. >4 n/wk preop: 18%; postop: 27%, p<.01, 24%, 9%, 15%, 22%, 29% >80% reported participation in recreational sports

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Panzram et al. 2018	Germany	Medial UKA	Retrospective cohort study	Patients: 27 Age: 62.5±8.3 (49-76) 56% male 11% bilateral	60±8.3 (47-69) months	Activity Sports participation	UCLA Tegner Schulthess Clinical Activity Questionnaire	Activity (1-10) High, medium, low (>6, 4-6, <4) Activity (1-10) Participation (%), RTS (%) RTS time Sport type	Preop: 4.9±2.3; postop: 6.1±1.8, p<.05 37%, 51%, 11% Preop: 2.9±1.4; postop: 3.4±1.0 89%, 100% 67% returned within 3 months Cycling, hiking, long walks most-performed. Impact sports decreased, barely different participation in low-impact sports
Panzram et al. 2020	Germany	UKA	Retrospective case series	Patients: 177 Age at FU: 64.4±9.7 (38-82) BMI at FU: 30.9±5.4 8% bilateral	37.1±9.8 (24-60) months	Activity Sports participation	UCLA Tegner Schulthess Clinical Activity Questionnaire	Activity (1-10), UCLA ≥7 (%) Activity (1-10) Sports participation (%) RTS (%), STS (%) Disciplines (n) Frequency (n), >4x/wk ≥ 60 min (%) Sport type	Preop: 2.9±1.7 (1-10); postop: 6.3±1.4 (2-10), 53.7% 3.5±1.0 (0-7) 83.4% 92.9%, 35.7% Preop: 2.8±2.2; postop: 2.0±1.8 Preop: 2.8±1.3; postop: 2.8±1.4 15.6% Low numbers in high-impact sports; cycling & walking most-performed
Paxton et al. 2016	United States	THA	Cohort study	Patients: 5678 Age: 67 [60-75] 41% male 62% BMI <30 0% bilateral	1-2 years	Activity	Questions by nursing staff	Activity (min/wk) Meeting guidelines	Preop: 50 [0-140]; postop: 150 [60-280] 50%
		TKA	Cohort study	Patients: 11084 Age: 68 [62-75] 38% male 49% BMI <30 0% bilateral	1-2 years	Activity	Questions by nursing staff	Activity (min/wk) Meeting guidelines	Preop: 58 [3-143]; postop: 150 [60-280] 50%
Payo-Ollero et al. 2020	Spain	THA	Case series	Patients: 46 Age: 41 (37-48) 72% male BMI: 26.1 (24.5-29) 25% bilateral	7.5 (1-11) years	Activity Sports participation	UCLA Phone questionnaire	Activity (1-10) Sports participation (%) Number of sports, 1, 2, 3, ≥4 (%) Sport type	Preop: 6.85±3.01; postop: 6.22±2.24 Preop: 74%; postop: 78% Preop: 20%, 24%, 11%, 20%; postopp: 28%, 28%, 11%, 11% Swimming, walking, cycling most-often performed
Pietsch-mann et al. 2013	Germany	Medial UKA	Case series	Patients: 131 Age: 65.3 (44-83) 44% male 6% bilateral	4.2 (1-10) years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Participation (%) RTS (%) Frequency Disciplines Impact Sport type	6 Preop: 60%, postop: 53% 80.1%, 11% of non-sporters started Increase in frequency Preop: 1.73, postop: 1.74 Shift from high- to low-impact (7 participated in impact sport preop, 3 postop) Cycling most popular activity
Pioger et al. 2020	France	TKA	Cross-sectional study	Patients: 143 active golfers Age: 65.7±(43-85) 85.3% male 0% bilateral	4.6 (2.0-14.2) years	Activity Golf	UCLA Questionnaire	Activity (1-10) Time to RTS (18-hole golf course), (months) Playtime (hrs/wk) Sport level (handicap)	Preop: 7.02±2.04; postop: 7.85±1.14; p< .01 3.7 (0.7-2.5) Preop: 8.9±6.1; postop: 10.2±6.6; p<.01 Preop: 25.2±13.1; Postop: 24.4±13.3; p=.01

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Pisanu et al. 2020	Italy	TKA PS (89%) TKA CNH (5%) TKA CR (3%) UKA (4%)	Retrospective case series	Patients: 118 Age: 73 [67-77] 28.7% male BMI: 29.1±4.7 3% bilateral	3.1±1.1 years	Sports participation	Phone questionnaire	Participation (%) RTS, STS (%) Time to RTS, (months) Disciplines (n) Activity type	87%, 93%, 50% 4 [3-6] Preop: 1.1; postop: 1.1 Most performed low-impact activities, mainly walking. No decrease in high- or medium-impact sports
Plassard et al. 2020	France	TKA	Retrospective case series	Patients: 443 Age: 69 (41-90) 36.6% male BMI: 29.3 (19-46)	43 [23-49] months	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Activity (%), low: UCLA ≤3, moderate 4-6, high ≥7 Sports participation (%) Sport type	Preop: 4.45; postop: 5.92 Preop: 27%, 64%, 8% Postop: 12%, 55%, 33% 85% Walking, hiking, swimming, hiking most-performed
Plate et al. 2013	United States	HRA (33%) CoC THA (33%) MoP THA (33%)	Therapeutic study	Patients: 3x30 matched groups Age: HRA & MoP THA: 53 (37-79) CoC THA: 52 (39-69) 37% male	HRA: 38 (24-66) MoP THA: 29 (24-60) CoC THA: (24-62) months	Activity	Weighted activity score questionnaire	Activity participation at recreational or professional level (0-9 low, >9 high)	HRA: 10.5 (1-28) CoP THA: 5.6 (1-18) CoC THA: 6.9 (0-34) HRA significantly higher than CoP and CoC THA.
Ponzio et al 2018	USA	TKA	Matched case-control study	Patients:1008 LEAS >12 preop (N=772 at FU) Age: 66.3±9.1 56.6% male BMI: 28.3±5.0	2 years	Activity	LEAS	Activity (1-18)	Preop: 14.6±1.2 Postop: 13.7±2.7
				Patients: 1008 LEAS 7-12 preop (N=727 at FU) Age: 66.3±9.0 56.6% male BMI: 28.4±4.9	2 years	Activity	LEAS	Activity (1-18)	Preop: 9.1±1.7; postop: 11.6±2.9
Poortinga et al. 2014	The Netherlands	THA (64%) TKA (36%)	Retrospective cohort study	Patients: 658 Age: 68±10.6 32% male BMI: 28.7±4.9	1 year	Activity	SQUASH	Total PA (min/wk) Leisure-time PA (min/wk) Meeting guideline (%)	Preop: 1140 [975-2040]; postop: 1525 [1200-1980]; p=.118 270 [120-570]; postop: 420 [210-720]; p<.001 Preop: 29%; postop: 40%; p=.005
Postler et al. 2017	Germany	THA	Comparative study	Patients: 124 Age: 64.1±11.0 29.8% male BMI: 27.5±3.7	2.3±0.3 years	Activity	UCLA	Activity (1-10)	Preop: 4.2±1.4; postop: 5.2±1.7
Pritchett 2018	United States	Tripolar THA	Prospective cohort study	Patients: 160 needing unrestricted activity Age: 43 (19-76) 48% male	11 (6-19) years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Sport type	Preop: 4 (2-7); postop: 8.8 (6-10), p<.001 80% UCLA ≥8 Many high-impact activities or participation at high level stated

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Ristolainen et al. 2019	Finland	TKA	Retrospective cohort study	Patients: 70, knee injury as cause of TKA Age: 51.8±5.2 BMI: 29.0±5.0 Sports-related: 18 Age: 50.0±5.3 39% male BMI: 27.8±4.1 44% bilateral Other causes: 45 Age: 52.2±5.3 56% male BMI: 29.9±5.5 29% bilateral	All: 6.2±3.7 years Sports-related: 4.5±3.0 years Other: 6.6±3.7 years	Activity Sports participation	Mailed questionnaire Mailed questionnaire	Leisure-time PA (MET hrs/wk) Participation Intensity Session duration Frequency	Sport: 42.1; Other: 18.5 Sport: preop: 100%, postop: 89% Other: preop: 71%, postop: 74% Total: 68% walking intensity, 14% high-intensity activity. Sports group participated more often at higher intensity >50% patients 1 hour >50% exercised >10x/month
Robertson et al. 2016	United States	THA (35%) TKA (15%) HRA (50%)	Case series	Patients: 13 active Age: 48.7 (37-61) 77% male BMI: 23.9 (20.6-28.5) 54% bilateral	8.7 (1.8-15.8) years	Activity	Accelerometer	Cycles/day Maximum cycles/day (over mean of 9.4 days) Cycles/year High PA cycles (%) Moderate PA cycles (%) Low PA cycles (%) High PA (%), (min/day) Moderate PA (%) (min/day)	8273 (5973-12557) 11794 (7925-16983) 3.019.737 (2.176.704-4.583.461) 39.7 (8-62) 42.6 (27.1-63.9) 18.7 (10.7-30.6) 4.3%, 58 (11.5-118.1) 9.4%, 135 (83.5-204.5)
Rolving et al. 2013	Denmark	THA	Historical cohort study	Patients: 95 Age: 72.3±6.2 33% male	22.4 (18.4-23.9) months	Activity Activity/ Sports participation	Physical activity scale Questionnaire	Physical activity scale (MET/day), >40 moderate Time spent on leisure activities (min/wk) Activity type	41 [38.5; 48.5] 480 [240; 870] Walking (79%), bicycling (48%)
Rozenlund et al. 2017	Denmark	THA	Randomized controlled trial	Patients: Lateral approach: 38 Age: 60±7 68% male BMI: 27±3 Posterior approach: 39 Age: 62±6 67% male BMI 28±4	12 months	Activity	UCLA	Activity (1-10)	Lateral approach: Preop: 5 [4-7]; postop: 6.5 Posterior approach: Preop: 5 [3-6]; postop: 7.3

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Rueckll et al. 2020	United States	THA	Prospective study	Patients: 39, desired UCLA ≥8 Age: 48.3±6.1 100% male BMI: 27.5±3.7 18% bilateral	56.7±13.9 months	Activity	UCLA LEAS	Activity (1-10) Activity (1-18)	Preop: 5.8±2.4; postop: 7.7±2.0 14.1±2.5
		HRA	Prospective study	Patients: 34, desired UCLA ≥8 Age: 47.6±5.2 100% male BMI: 27.6±3.3 3% bilateral	54.4±21.3 months	Activity	UCLA LEAS	Activity (1-10) Activity (1-18)	Preop: 6.0±2.3; postop: 9.4±2.1 15.9±1.5
Sandberg et al. 2019	United States	TKA	Matched case-control study	Patients: 183 Age: 72.1±7.8 27.9% male BMI: 33.1±6.4	12.9 months	Activity	UCLA	Activity (1-10)	TKA: Preop: 3.90±1.57; postop: 5.12±1.68 6.2% UCLA >7
		Dual-pivot TKA	Matched case-control study	Patients: 183 Age: 67.8±7.9 27.2% male BMI: 33.4±6.5	12.6 months	Activity	UCLA	Activity (1-10)	Dual-pivot TKA: Preop: 4.03±1.72; postop: 5.31±1.85 12.2% UCLA >7 No significant differences between standard TKA and dual-pivot TKA
Sandiford et al. 2015	United Kingdom	HRA	Case series	Patients: 79 active Age: 54.9 (34.5-73.6) 67% male BMI: 25.2 (19.8-31.9) 1% bilateral	8.5 (8-10) years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) RTS (%) RTS time Disciplines (n), >1 (%) Sport type Sport level Frequency (n/wk) Session duration (min)	Preop: 4 (1-9); postop: 7.6 (2-10) 6 months: 100%, 8.5yr: 97% 85% returned within 3 months, 100% within 6 months 2 (1-5): 37% performed more disciplines postop. Walking, swimming. aerobics most-frequently performed 14% competitive, 86% recreational All >3x/wk All 60-90
Schmidutz et al. 2012	Germany	Short-stem THA	Case series	Patients: 68 Age: 55±12 Gender: 60% male BMI: 26±4 kg/m2 12% bilateral	2.7±0.7 (2.0-4.2) years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) Participation, RTS (%) RTS time (1-2, 3-4, >5 mo) Disciplines (n) Sport impact Sport type Frequency, (n/wk) Session duration (min)	7.6±1.9 (3-10) 91%, 98% 26%, 25%, 47% Preop: 3.9±2.4; 3.5±2.0 41% fewer disciplines, 59% equal or more disciplines Decrease in high- and medium-impact sports, 41% participating in high- or medium-impact activities Cycling, hiking, swimming most popular Preop: 1.5±0.9; postop: 1.8±1.1; p=.022 Preop: 67±35; postop: 66±33; p=.753
Schneider et al. 2020	United States	TKA (50%) medial UKA (32%) lateral UKA (19%)	Retrospective cohort study	Patients: 92 did sports preop Age: 56±9.1 13% male BMI: 28±5.6	1-2 years	Activity Sports participation	UCLA Questionnaire	Activity (1-10) UCLA ≥7 (%) Sport impact (low, medium, high), (number of sports)	TKA: Preop: 5.26±2.4; postop: 6.62±1.9 Preop: 39.2%; 60.8% 2.61±1.6; 1.00±0.9; 0.33±0.7
						Activity Sports participation	UCLA Questionnaire	Activity (1-10) UCLA ≥7 (%) Sport impact (low, medium, high), (number of sports)	UKA: Preop: 5.93±2.5; postop: 6.89±2.0 Preop: 32.6%; 54.3% 2.82±2.0; 1.07±1.0; 0.48±0.7

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Scott et al. 2017	United Kingdom	TKA	Case series	Patients: 289 Age: 59 (42-65) 49% male BMI: 31.4 (15-55) 8% bilateral	3.3 (2-4) years	Activity	UCLA	Activity (1-10) (Median)	Preop: 4 Postop: 6 58% improved 46% ≥7
Smith et al. 2017 (T)	United Kingdom	THA (57%) TKA (43%)	Prospective cohort study	Patients: 201 Age: 72.9±10.7 38% male 0% unilateral	2-4 years	Activity	Face-to-face interviews	Summary index (sedentary, mild, moderate, high)	Preop: 7.5% (5-12); 17.4% (13-23); 61.7% (55-68); 13.4% (9-18) Postop: 12.9% (9-18); 20.4% (15-27); 54.7% (48-61); 11.9% (8-17)
Smith et al. 2018 Does PA change	United Kingdom	THA TKA	Prospective case-control study	THA patients: 105 Age: 68.2±9.3 42.9% male BMI: 28.8±4.2 TKA patients: 116 Age: 67.3±8.3 41.4% male BMI: 30.1±4.9	12 months, 24 months	Activity Sports participation	PASE PASE	Total score Light sport/ recreation frequency (never, 1-2, 3-4, >4 days) Moderate sport/ recreation frequency (never, 1-2, 3-4, >4 days) Strenuous sport/ recreation frequency (never, 1-2, 3-4, >4 days)	THA preop: 136; 12mo: 135; 24mo: 132 TKA preop: 126; 12mo: 121; 24mo: 142 TKA preop to 24 mo significant increase No differences with control group THA 24 mo: 87.5%, 9.7%, 2.8%, 0% TKA 24 mo: 84.5%, 13.6%, 1.0%, 1.0% THA 24 mo: 94.4%, 2.8%, 1.4%, 1.4% TKA 24 mo: 87.4%, 6.8%, 4.9%, 0.9% THA 24 mo: 73.6%, 4.2%, 13.9%, 8.3% TKA 24 mo: 66%, 8.7%, 18.4%, 6.9%
Stambourgh et al. 2014	USA	TKA	Matched case-control study	Patients: 76 Age: 48.7 (26-55) 30% male BMI: 33.9 (14.9-54.9) 7% bilateral	Mean FU 4.8 (2-13.9) years	Activity	UCLA	Activity (1-10)	Preop: 3.47 (1-9) Postop: 4.71 (2-10)
Takenaga et al 2013	USA	THA	Cohort study	Patients: 55 Age: 39 (18-50) 65% male BMI: 29 (18-42) 5% bilateral	12.1 (10-16) years	Activity	UCLA Tegner Step Activity Monitor	Activity (1-10) Activity (1-10) Steps/year	6.1 (2-10) 3 (0-6) 1.56 million steps (77,000-3.26 million)
Takeuchi et al. 2020	Japan	THA	Retrospective case series	Patients: 204 Age: 53.7 (30-60) 18% male 18% bilateral	59.5 months	Activity	UCLA	Activity (1-10)	Preop: 4.55; postop: 6.17; p<.01 Preop highest UCLA was 6, achieved by 16.7%, but postop 62.3% had UCLA 6 and 18.6% >7
Vielgut et al. 2016	Austria	TKA	Retrospective case series	Patients: 236 Age: 62.7±11.4 18% male 10% bilateral	14.9±3.0 years	Activity Sports participation	Tegner score	Activity (1-10) Participation, RTS (%) Sport impact Sport type	Preop: 2.18±1.5; postop: 3.04±1.5 70.8%, 70.8% 71.3% continued participating in low-impact, 43.7% medium-impact, 16.7% high-impact sports Low-impact activities most-performed, especially cycling & hiking.
Von Rottkay et al. 2018	Germany	THA	Prospective cohort study	Patients: 64 Age: 63.0 53% male BMI: 27.1	12 months	Activity	Daily activity questionnaire Step counter	Load cycles/day Steps/day	Preop: 3727±947; postop: 4226±948, Preop: 5541±2048; postop: 5658±2213

Study	Country	Arthroplasty	Study design	Sample size and Characteristics*	Data collection period (FU & time postop)	Type of sport	Measurement method	Outcome variables of interest	Outcomes
Wagenmakers et al. 2011	The Netherlands	THA	Prospective multicentre cohort study	Patients: 653 Age: 70.3±8.2 26% male BMI: 27.0±4.1	52.4±3.9 weeks	Activity	SQUASH	Activity (min/wk) LPA (min/wk) MPA (min/wk) VPA (min/wk) Meeting guidelines (%)	1468.1±1138.3 Most time spent in household (756.3±766.5) and leisure (584.5±657.8) activities 805.1±800.4 333.6±508.0 329.4±458.5 67%
Walker et al. 2015a (return to sports)	Germany	Lateral UKA	Retrospective case series	Patients: 45 Age: 60.1±10.5 42% male BMI: 27 0% bilateral	3 (2.0-4.3) years	Activity Sports participation	UCLA Tegner score Schulthess Clinic Activity score	Activity (1-10), ≥7 (%) Activity (1-10) Participation, RTS (%) RTS time (<3, <6, >6 mo) Disciplines (n) Frequency (n/wk) >1-hr session (%) Sport impact Sport type	Preop: 5.3±2.3; postop: 6.7±1.5, 66% Preop: 2.6±1.6; postop: 3.5±0.8 Preop: 93%; postop: 95.6%, RTS: 97.8% 55%, 77.8%, 22.2% Preop: 3; postop: 3 Preop: 3; postop: 3 Preop: 52%; postop: 44% Decrease in high-impact activities, increase in low-impact activities Biking, walking/hiking most performed
Walker et al. 2015b (Sports, PA)	Germany	Medial UKA	Retrospective case series	Patients: 93 Age: 55 (36-60) 48% male BMI: 32 (20-58) 17% bilateral	4.4±1.6 (2.3-8.4) years	Activity Sports participation	UCLA Tegner score Schulthess Clinic Activity score	Activity (1-10), ≥7 (%) Activity (1-10) Participation (%) RTS; STS (%) RTS time (<1, <3, <6, >6 mo) Disciplines (n) Frequency (n/wk) >1-hr session (%) Sport impact Sport type	Preop: 3.3±1.5; postop: 6.8±1.5; 62% Preop: 2.0±1.1; postop: 3.8±1.1 Preop: 93%, postop: 91%; 93%; 6% 27%; 56%; 77%; 23% Preop: 4, postop: 3.3 53% participated in ≥3 sessions/wk 45% Decrease in high-impact activities, increase in low-impact activities. 29% participated in high-impact sports Biking, walking, swimming most-performed
Webber et al. 2017	Canada	TKA	Cross-sectional study	Patients: 38 Age: 67.9±7.3 42% male BMI: 30.5±6.1	1 year	Activity	Activity monitor	Steps (steps/day) Sedentary, (hrs/day); (%/wear time) LPA (min/day) MVPA (min/day)	Preop: 3724±2338; postop: 5935±3316 Preop: 9.3±1.4, postop: 9.2±1.4, p=.62 Preop: 66.9%±9.0, postop: 63.8%±10.0; p=.18 Preop: 264.0±73.3; postop: 291.7±91.0 Preop: 0±1.8; postop: 2.1±10.4 Patients had a higher cadence postop
Wimmer et al. 2015	United States	TKA	Cross-sectional cohort study	Patients: 32 Age: 77.8±6.1 31% male BMI: 29.9±5.7	11.5±3.3 years	Activity	Activity monitor and goniometer	Lying (%) Sitting (%) Standing (%) Level-walking (%) Stair-walking (%) Unrecognized (%) Steps (steps/12hrs)	7.8%±10.5 59.9%±14.7 12.7%±5.8 7.8%±3.8 0.1%±0.1 11.7%±3.6 3102±1553, of which 2.1% stairs
Wollmerstedt et al. 2010	Germany	THA	Cross-sectional cohort study	Patients: 59 Age: 58 (37-77) 50% male	5 years	Activity	Pedometer Daily Activity Questionnaire (DAQ)	Steps (steps/day) Steps (steps/day)	Preop: 4967±2154; 5yr: 6564±2795 Preop: 4665±2053; 5yr: 6375±2971
				Patients: 75 Age: 70 (52-86) 48% male	10 years	Activity	UCLA Pedometer Daily Activity Questionnaire (DAQ)	Activity (1-10) Steps (steps/day) Steps (steps/day)	Preop: 6±1; postop: 6±1 5101±2412 5210±2738

Zimmerer et al. 2021	Germany	Lateral UKA	Case series	Patients: 19 Age: 26.7±10.5 (36.0-76.0) 26% male BMI: 27.2±5.7 (17.3-38.0)	4.6±2.2 (2.0-9.7) years	Activity Sports activity	UCLA Schulthess Clinic Activity Score	Activity (1-10) UCLA ≥7 (%) Participation (%) RTS (%), STS (%) Disciplines (n) Sport frequency (n/wk) Session length (min) Sport type	Preop: 4.1±1.7 (2-9); postop: 6.4±1.3 (4-9) 47% 68% 86%, 0% Preop: 2.8; postop: 2.6 Preop: 2.7; postop: 2.3 Preop: 41.5; postop: 45.9 Long walks & biking most-performed. High-impact sports (soccer, skiing) had decreased participation rates postop. Nordic walking & jogging significantly increased.

*At time of surgery, unless indicated otherwise; ? Unknown/not reported.
Abbreviations: BFH: big femoral head; BMI: body mass index; CNH: constrained non-hinged; CoC: ceramic-on-ceramic; CoP: ceramic-on-polyethylene; CR: cruciate-retaining; FU: follow-up; HPA: high physical activity; HRA: hip-resurfacing arthroplasty; IPAQ: International Physical Activity Questionnaire; IQR: interquartile range; LEAS: Lower Extremity Activity Scale; LPA: low physical activity; MET: metabolic equivalent of task; MoM: metal-on-metal; MoP: metal-on-polyethylene; MPA: moderate physical activity; MVPA: moderate-to-vigorous physical activity; PA: physical activity; PASE: Physical Activity Scale for the Elderly; postop: postoperatively; preop: preoperatively; PS: posterior stabilized; RTS: return to sports; SMF: short metaphyseal; SQUASH: Short QUestionnaire to ASsess Health-enhancing physical activity; STS: start to sport; THA: total hip arthroplasty; TKA: total knee arthroplasty; UCLA: University of California Los Angeles activity scale; UKA: unicompartimental knee arthroplasty; VPA: vigorous physical activity; YPAS: Yale Physical Activity Scale

Grimby scale (range 1-4) 1 sedentary – 4 regular hard physical training for competition sports (vigorous physical activity; LEAS: Lower Extremity Activity Scale (range 1-18) 1 I am confined to my bed all day – 18 I am up and about at will in my house and outside. I also participate in vigorous physical activity such as competitive level sports daily; Sports activity index (range 0-25) 0-6 = non-active; 7-10 = moderately active; 11-25 = active; Tegner activity scale (range 0-10) 0 is on sick leave/disability - 10 is participation in competive sports at a national or international elite level; UCLA: University of California Los Angeles activity scale (range 1-10) 1 wholly inactive: dependent on others; cannot leave residence – 10 regular participation in impact sports such as tennis, acrobatics, ballet, heavy labour or backpacking; Weighted activity score (0-9 low, >9 high)